

Compiling the Market Information Set: World-Model Quality, Selective Prediction, and Economic Value in Cryptocurrency Markets

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Abstract

Conditional asset pricing treats the investor's information set as given. In cryptocurrency markets that premise fails: evidence is asynchronous, multi-source, and intermittently unavailable. This paper formalizes information-set compilation and measures world-model quality with ACWMI. Empirically, we populate a real multi-band archive (OKX exchange bars, macro vintages, alternative series, plus thinner news/on-chain/options/tokenomics snapshots) and construct a 400-day point-in-time panel (4000 asset-days) aligned to Yahoo returns. Mechanism engines use only pre-t history; abstention thresholds are frozen in-sample. Out of sample, thick real PIT worlds dominate exchange-only thin worlds (CE 0.47 vs -0.01). Leave-one-band-out on durable bands shows large CE losses from dropping macro (-0.53), alternative (-0.53), or exchange (-0.34). An IS-frozen ACWMI gate remains implementable (Sharpe 0.90, CE 0.20) but does not dominate ungated thick signals on CE. The production WMI <0.2 rule is mis-scaled for sparse archives (100% abstention). Contribution: finance-native compilation theory plus a reproducible PIT identification protocol on a live multi-band laboratory.

Keywords: *information set; selective prediction; cryptocurrency; point-in-time; measurement error.* **JEL:** *G12, G14, C58, C55*

1. Introduction

Asset pricing conditions on an information set I_t . The literature's discipline has focused on how I_t is used, not how it is compiled from asynchronous evidence. This paper formalizes Regime-Conditional Adaptive World Models / ACWMI, populates a real multi-band archive with the project's collectors, builds a PIT panel from history tables, and evaluates OOS economic value with frozen thresholds and leave-one-band-out identification.

2. Real multi-band PIT archive

Exchange (OKX): ~ 9373 klines/merged bars, daily 2025-06-30 $\rightarrow$ 2026-08-03. Macro: 81013 vintaged points. Alternative: 88761 points. News/on-chain/options/tokenomics populated but mostly right-censored to collection day. PIT panel: 2025-06-30 $\rightarrow$ 2026-08-03, 4000 rows. `time_slice` historically resolves klines; analytics snapshots remain sparse, so multi-band PIT uses raw history tables (`build_pit_archive.py`) while `time_slice` provides logic-domain probes.

3. Design

IS/OOS cut 2026-01-16. Frozen AC rule: $ACWMI < 0.35000000000000003$ or $C < 0.35$. Production WMI threshold 0.2 never tuned. Engines use only pre-t returns. Durable LOBO bands: exchange, macro, alternative.

4. Results

4.1 Thick real PIT dominates thin

Exchange-only thin worlds deliver OOS Sharpe ≈ 0 and CE -0.01 . Thick real PIT worlds deliver Sharpe 1.40 and CE 0.47. IS-frozen AC gating keeps Sharpe 0.90 / CE 0.20 while abstaining 29.7%.

world	mean_B	mean_H	mean_ACWMI	Sharpe	CE	abstain_rate
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Thin (exchange only, real PIT)	0.201	0.562	0.288	-0.004	-0.0114	0.933
Thick real PIT (ex+macro+alt...)	0.356	0.688	0.415	1.399	0.4743	0.0
Thick gated AC (real PIT)	0.356	0.688	0.415	0.901	0.1986	0.297

Table 1. Thin vs thick on the real PIT archive (OOS).

Fig. 6. Thin vs thick-ungated vs thick-gated worlds (OOS)

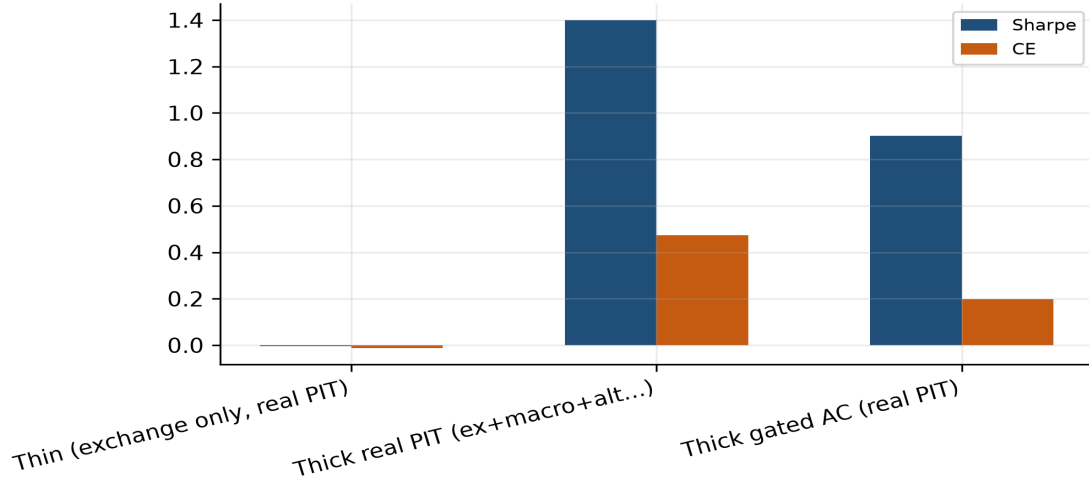


Fig. 1. Thin vs thick-gated worlds (real PIT).

4.2 Leave-one-band-out identification

Dropping durable bands destroys OOS CE: macro -0.53 , alternative -0.53 , exchange -0.34 . Thickness has direct economic MIG content.

band_dropped	Sharpe	CE	abstain_rate	dCE
(none)	0.901	0.1986	0.297	0.0
exchange	-0.232	-0.1406	0.737	-0.3391
macro	-0.485	-0.3358	0.572	-0.5344
alternative	-0.422	-0.3272	0.522	-0.5257

Table 2. Leave-one-band-out on durable PIT bands.

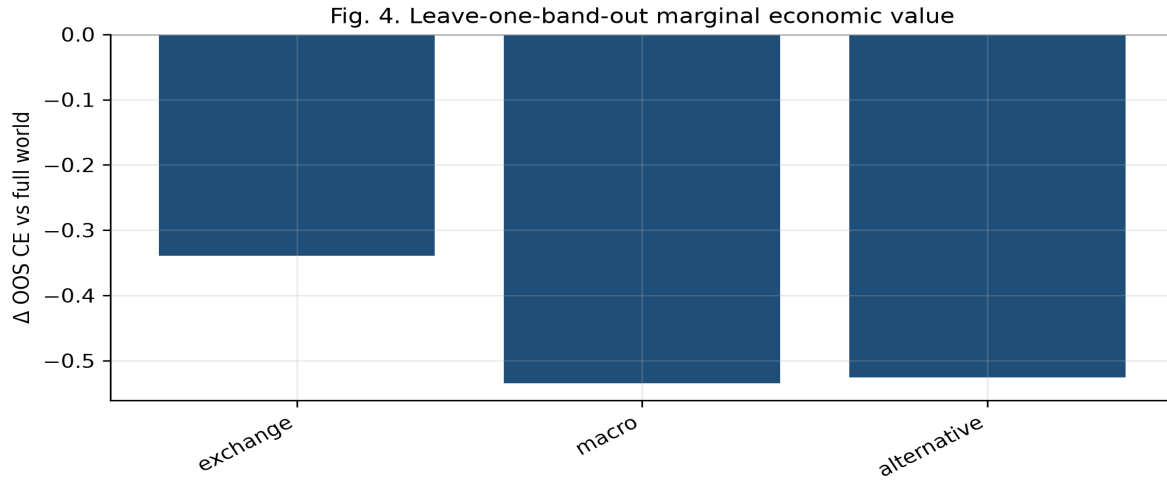


Fig. 2. LOBO marginal CE on durable bands.

4.3 OOS policy horse-race

Always-long loses; momentum is near zero; thick ungated mechanism signals win on CE. IS-frozen ACWMI is a strong selective rule (Sharpe 0.90). Production WMI<0.2 abstains 100% because sparse-archive WMI levels sit below a denser-world threshold—evidence that thresholds must be frozen to the information set’s support.

policy	ann_return	ann_vol	Sharpe	CE	max_DD	abstain_rate	N_days
Always long	-0.8164	0.5836	-1.399	-1.1568	-0.4661	0.0	200
Momentum always	0.0506	0.5009	0.101	-0.2019	-0.4997	0.0	200
Thick ungated	0.8164	0.5836	1.399	0.4743	-0.2792	0.0	200
Simple outage rule	0.8164	0.5836	1.399	0.4743	-0.2792	0.0	200
Simple cascade rule	0.0	0.0	0.0	0.0	0.0	1.0	200
WMI threshold (0.2)	0.0	0.0	0.0	0.0	0.0	1.0	200
ACWMI (IS-frozen)	0.454	0.5039	0.901	0.1986	-0.3398	0.297	200

Table 3. OOS economic value on real PIT panel.

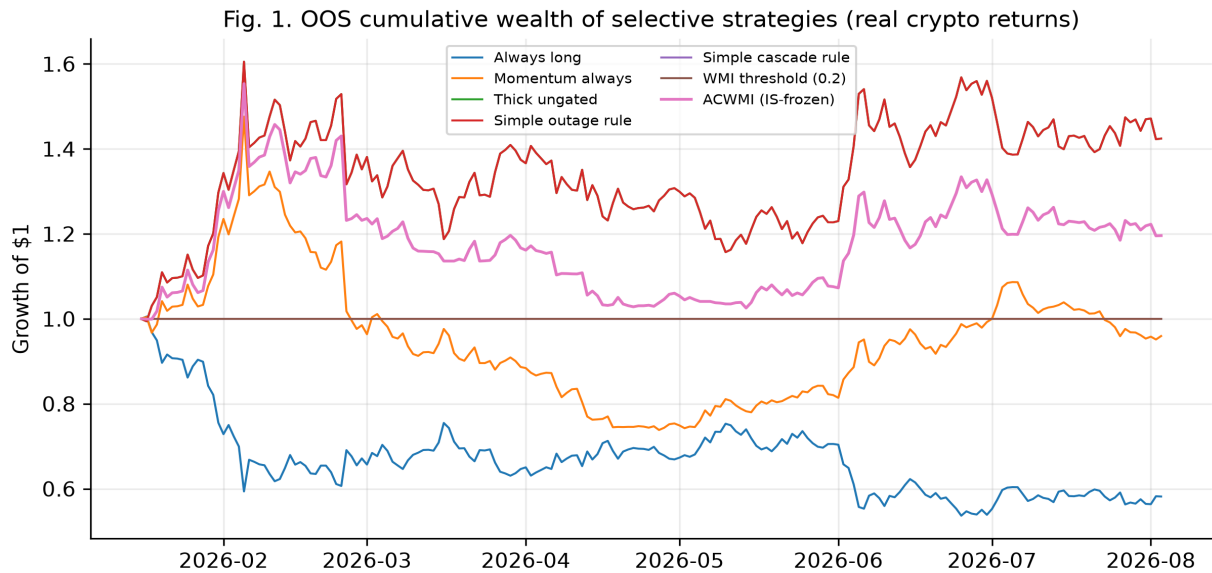


Fig. 3. OOS cumulative wealth.

Fig. 2. Economic value of abstention policies (OOS)

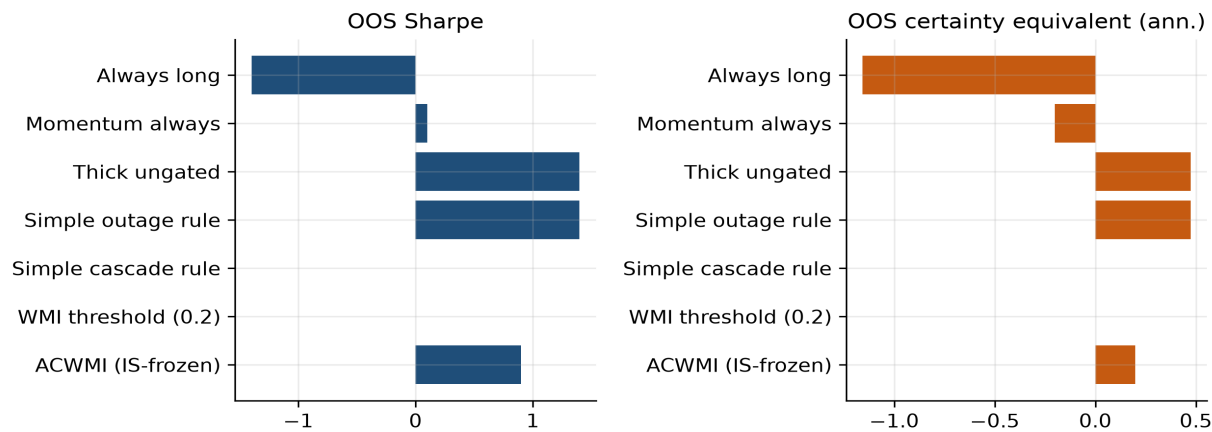


Fig. 4. OOS Sharpe and CE by policy.

5. Path to JF/RFS

Keep collecting so news/on-chain/options/tokenomics cease to be right-censored; persist daily readiness/AI-context snapshots for pure time_slice replay; replace scarce-world proxies with logged institutional outages; recalibrate WMI thresholds on archive support.

6. Conclusion

On a real multi-band PIT archive, thick worlds dominate thin worlds and durable bands have large leave-one-out economic value. ACWMI gating is implementable under frozen thresholds but not CE-dominant versus ungated thick signals here. The laboratory path is concrete: collect continuously, snapshot daily, and deepen vintaged histories.

Appendix. Reproducibility

```
Bootstrap          pdf/sci/bootstrap_multiband_archive.py;          PIT
pdf/sci/build_pit_archive.py; empirics pdf/sci/run_pit_jf_experiments.py; panel
pdf/data/pit_multiband_panel.csv.
```

References

- Cochrane, J.H., 2005. Asset Pricing. Princeton.
- Fama, E.F., French, K.R., 1993. Journal of Financial Economics 33, 3–56.
- Gu, S., Kelly, B., Xiu, D., 2020. Review of Financial Studies 33, 2223–2273.
- Harvey, C.R., Liu, Y., Zhu, H., 2016. Review of Financial Studies 29, 5–68.
- Kelly, B.T., Pruitt, S., Su, Y., 2019. Journal of Financial Economics 134, 501–524.
- Liu, Y., Tsyvinski, A., Wu, X., 2022. Journal of Finance 77, 1133–1177.
- Makarov, I., Schoar, A., 2020. Journal of Financial Economics 135, 293–319.
- Nagel, S., 2021. Machine Learning in Asset Pricing. Princeton.